

Backprop & Gradient Descent

Loss is a landscape. Every weight is a dimension. The gradient points downhill. Training is an iterative ball-rolling procedure — nothing more. Today we make that intuition rigorous.

Before we start — guess how many gradient steps OpenAI ran to train GPT-4. Reveal at minute 45.

WHAT'S INSIDE

- 01** Gradient descent — a ball downhill
- 02** Chain rule — multiply the slopes
- 03** One step by hand
- 04** Optimisers — one-line comparison
- 05** Read the loss curve — the one diagnostic

WHY THIS LESSON MATTERS

A network does not "learn". It rolls downhill.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Explain gradient descent as iterative error reduction.

2

Compute the gradient of a scalar loss by hand.

3

Describe the chain rule and why backprop is efficient.

4

Compare SGD, Momentum, RMSProp, Adam conceptually.

5

Diagnose a learning rate from the loss curve.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

**Gradient descent —
a ball...**

Loss landscape · gradient · step size

02

**Chain rule —
multiply the...**

Why "back" propagation

03

One step by hand

Tiny network: $x \rightarrow w \rightarrow \hat{y}$; $\text{loss} = (\hat{y} - y)^2$

04

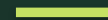
**Optimisers — one-
line...**

Start with Adam at $\text{lr}=1\text{e-}3$. Tune only when you see a concrete problem.

01

PART 1 OF 4

Gradient descent — a ball...



Loss landscape · gradient · step size

CONCEPT 1 OF 4

Gradient descent — a ball downhill

Loss: a single scalar measuring model error.

Gradient: $\partial \text{Loss} / \partial \text{parameter}$ — direction of steepest ascent.

Descent: take a small step opposite the gradient.

Learning rate: the step size. Too big → bounce. Too small → glacial.

GOAL

Minimum loss

Find the lowest point on a landscape you cannot see.

DEFINITION

Gradient descent — a ball downhill

Loss landscape · gradient · step size

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Chain rule — multiply the slopes

$$y = f(g(x)) \rightarrow dy/dx = df/dg \cdot dg/dx.$$

For a deep network, each weight's gradient = product along the path.

Backprop computes ALL gradients in ONE backward sweep.

Without this efficiency, modern deep learning is intractable.

MIRACLE

One backward sweep

Linear cost in parameters — not exponential.

KEY POINTS

- $y = f(g(x)) \rightarrow dy/dx = df/dg \cdot dg/dx$
- For a deep network, each weight's gradient = product along the path
- Backprop computes ALL gradients in ONE backward sweep
- Without this efficiency, modern deep learning is intractable

CONCEPT 3 OF 4

One step by hand

Forward: $\hat{y} = w \cdot x$.

Loss: $L = (w \cdot x - y)^2$.

$dL/dw = 2 \cdot (w \cdot x - y) \cdot x$.

Update: $w \leftarrow w - \eta \cdot dL/dw$.

PRACTISE

Pen + paper

Do this once by hand; PyTorch becomes obvious.

DEFINITION

One step by hand

Tiny network: $x \rightarrow w \rightarrow \hat{y}$; loss = $(\hat{y} - y)^2$

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Optimisers — one-line comparison

Start with Adam at $lr=1e-3$. Tune only when you see a concrete problem.

KEY POINTS

- Start with Adam at $lr=1e-3$. Tune only when you see a concrete problem
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

One gradient descent step — paper only

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Why does the chain rule make deep learning possible?

Q2

Which optimiser will you default to — and why?

Q3

What shape of loss curve will you show your team when debugging?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

A gradient tells you:

A

Loss value

B

Direction of steepest descent

C

Learning rate

D

Model accuracy

A gradient tells you:



CORRECT ANSWER

Direction of steepest descent

WHY

It is the direction of steepest ASCENT; we step opposite to descend

Why "back" propagation?

A

Speed

B

Chain rule applied from loss backward

C

Memory

D

Tradition

Why "back" propagation?

CORRECT ANSWER



Chain rule applied from loss backward

WHY

Chain rule sweeps back from the loss, reusing intermediate gradients

Oscillating loss each epoch means:

A

LR too small

B

LR too big

C

Model too small

D

Data bug

Oscillating loss each epoch means:



CORRECT ANSWER

LR too big

WHY

Step size exceeds the local valley width; reduce LR

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Loss: a single scalar measuring model error.

Gradient: $\partial \text{Loss} / \partial \text{parameter}$ — direction of steepest ascent.

Descent: take a small step opposite the gradient.

Learning rate: the step size. Too big → bounce. Too small → glacial.

GOAL

Minimum loss

Find the lowest point on a landscape you cannot see.

THE TAKEAWAY

"Plot the loss. Read the curve. THEN tune."

WHAT TO NOTICE

1

Gradient descent — a ball...

2

Chain rule — multiply the...

3

One step by hand

4

Optimisers — one-line...

Mini assignment — learning-rate sweep

Train your Lesson-2 MLP with $\text{lr} = 1\text{e-}2$, $1\text{e-}3$, $1\text{e-}4$ for 20 epochs each.

DELIVERABLES

- Record loss curves for all three learning rates.
- Plot them on one axis.
- Write 3 sentences on what you observed.
- Push to GitHub.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Single plot + 3-sentence write-up on the forum.

ONE THING TO REMEMBER

“

"Plot the loss. Read the curve. THEN tune."

— AI Learning Hub

END OF LESSON 3

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 4 — NLP Fundamentals. Text → numbers → insight.